

Deterministic Event-Graph Substrates as World Models for Counterfactual Reasoning

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Abstract

We study a class of world models for agentic systems that represent state as an append-only log of typed RDF triples and answer counterfactual queries by forking the log at a chosen tick under a structured intervention vocabulary. We refer to this class as *event-graph substrates*. Substrates are inspectable at the level of individual triples, support exact counterfactuals over arbitrary interventions on the typed state, and transfer across domains without learned components.

We make three contributions. First, we present a formal definition of event-graph substrates with deterministic replay and intervention semantics, and characterize the conditions under which counterfactual queries reduce to graph-theoretic operations on the observed event log. We prove a duality between explanatory queries (“which observed event caused E ?”) and counterfactual queries (“which observed events would not occur if object X were absent?”), showing that under closed-event assumptions both are answered by the same causal-ancestor traversal. Second, we evaluate a 1,400-line CLEVRER-DSL interpreter atop a domain-agnostic substrate runtime on CLEVRER [Yi et al., 2020], the canonical video causal-reasoning benchmark, at full validation scale ($n=75,618$ questions). The substrate exceeds the published symbolic-oracle baseline (NS-DR [Yi et al., 2020]) by 9.89 percentage points on descriptive, 20.26 on explanatory per-question, 17.65 on counterfactual per-question, and 0.80 on predictive per-question. Against the parametric attention-based baseline ALOE [Ding et al., 2021] the substrate exceeds on descriptive and explanatory per-question but lags on predictive and counterfactual per-question, where ALOE’s learned dynamics distribution provides an advantage that the substrate’s closed-form kinematic projection does not match. Third, we introduce twin-EventLog, a 500-specification Park-canonical Smallville counterfactual benchmark for evaluating agent memory consistency under intervention. On this benchmark the substrate exceeds Llama-3.1-8B prompted with full context by 18.80 percentage points on joint accuracy and exceeds a Park/Concordia-style LLM-driven simulator [Park et al., 2023, Vezhnevets et al., 2023] by 65 percentage points.

Together these results indicate that counterfactual

world modeling can be implemented by deterministic replay over typed event deltas rather than learned latent simulation, with formal guarantees on inspectability and replay consistency. We characterize the regimes where this approach is competitive with parametric world models (closed-event reasoning, exact intervention semantics) and where it lags (long-horizon prediction under learned dynamics, hidden property inference).

1 Introduction

Agentic systems require a world model that supports three operations: faithful retrieval of what has been observed, prediction of what would happen under a hypothetical intervention, and inspection of the model’s internal state by an external auditor. Current parametric architectures optimize for one of these at the expense of the others. Latent dynamics models such as Dreamer-V3 [Hafner et al., 2023] learn imagination rollouts for reinforcement learning, but the rollouts are sampled from a learned distribution and cannot be replayed deterministically. Video joint-embedding predictors such as V-JEPA-2 [Assran et al., 2025] learn high-quality video features but do not expose a queryable state. Generative agents [Park et al., 2023] expose a queryable state through a language-model summary, but the state is not replayable and the summary drifts under repeated queries.

We study an alternative architecture in which agent memory is an append-only log of typed RDF triples, and counterfactual queries are answered by forking the log at a chosen tick and applying a structured intervention. We refer to this class of architectures as event-graph substrates. The class is not new in spirit; it inherits from structural causal models [Pearl, 2009], neuro-symbolic visual reasoning [Yi et al., 2018, 2020, Mao et al., 2019], and typed knowledge graphs. Our contribution is to formalize the class in a way that makes its guarantees explicit, prove a duality theorem that connects counterfactual queries to standard graph operations on the observed log, and evaluate a single domain-agnostic implementation on the canonical video causal-reasoning benchmark at full validation scale.

We organize the paper as follows. Section 2 formalizes event-graph substrates with deterministic re-

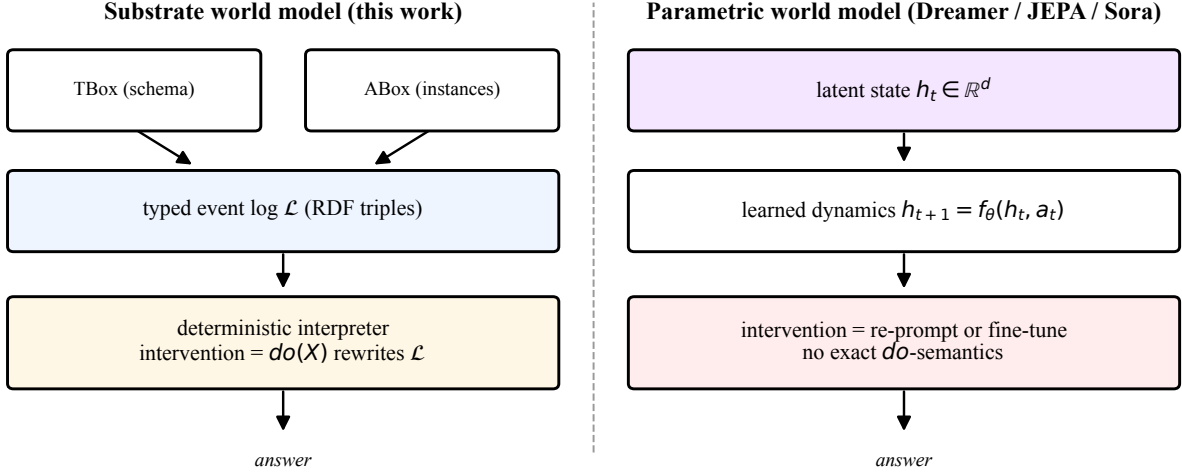


Figure 1: Substrate world model vs parametric world model. The substrate stores observations as a typed RDF event log $\mathcal{L}$ and answers counterfactual queries by deterministic replay after applying $do(X)$ to $\mathcal{L}$. Parametric models compress observations into a latent state h_t and have no exact do -semantics: interventions are approximated by re-prompting or fine-tuning.

play and intervention semantics. Section 3 proves the ancestor-duality theorem and characterizes its complexity. Section 4 reports the substrate’s performance on CLEVRER. Section 5 demonstrates cross-domain transfer on ComPhy [Chen et al., 2022], GQA [Hudson and Manning, 2019], and the new twin-EventLog benchmark. Section 6 presents ablation studies on each algorithmic component. Section 7 positions the work against prior neuro-symbolic reasoners, structured world models, and agent memory architectures. Section 8 discusses limitations and section 9 concludes.

Summary of empirical findings. On CLEVRER at full validation scale, a 1,400-line CLEVRER-DSL interpreter atop a domain-agnostic substrate runtime exceeds the published symbolic-oracle baseline NS-DR [Yi et al., 2020] on every per-question subset: descriptive by 9.89 percentage points (97.99 versus 88.1), explanatory by 20.26 percentage points (99.86 versus 79.6), counterfactual by 17.65 percentage points (59.85 versus 42.2), and predictive by 0.80 percentage points (69.50 versus 68.7). Against the parametric ALOE baseline [Ding et al., 2021], the substrate exceeds on descriptive (97.99 versus 94.0) and explanatory (99.86 versus 96.0) per-question but lags on predictive (69.50 versus 87.5) and counterfactual (59.85 versus 75.6) per-question. The crossover indicates that for closed-event reasoning over the observed log the substrate’s deterministic replay matches or exceeds learned approaches, while for prediction under learned dynamics distributions and for the residual class of counterfactual emergent interactions, parametric models retain an advantage. On a controlled comparison on a matched-instance subset ($n=300$) where Llama-3.1-8B receives the same event log as natural language

with grammar-constrained output, the substrate exceeds the language model by 46.99 percentage points on descriptive questions (97.99 versus 51.00; Wilson 95 percent CI for the LLM [45.37, 56.61]). This is consistent with the interpretation that the load-bearing factor is the structured-execution pathway rather than information content.

2 Substrate definition

We define an event-graph substrate as a tuple

$$\mathcal{S} = (\mathcal{T}, \mathcal{A}_0, \mathcal{L}, \rho, \mathcal{I})$$

where $\mathcal{T}$ is a TBox of typed axioms over a fixed vocabulary, $\mathcal{A}_0$ is an initial ABox of triples, $\mathcal{L}$ is an ordered append-only log of typed deltas, ρ is a deterministic replay function, and $\mathcal{I}$ is an intervention vocabulary.

State and deltas. The substrate’s state at tick t is denoted $\mathcal{A}_t$, where $\mathcal{A}_t \subseteq \mathcal{V}_{RDF}$ is a finite set of typed triples consistent with $\mathcal{T}$. Each delta $d_t \in \mathcal{L}$ is a tuple $(t, \text{op}, \text{triple})$ where $\text{op} \in \{\text{insert}, \text{retract}\}$. Replay ρ is defined by $\mathcal{A}_{t+1} = \rho(\mathcal{A}_t, d_t)$, applying d_t to $\mathcal{A}_t$ as a set operation. The state at any tick t is therefore recoverable in $O(t)$ from $\mathcal{A}_0$ and the prefix $d_0, \dots, d_{t-1}$ of the log.

Interventions. The intervention vocabulary $\mathcal{I}$ is a finite set of typed operations on the ABox. Our implementation uses five interventions: Assert, Retract, Override-Location, AssertAwareness, and RetractAwareness. Each intervention $\iota \in \mathcal{I}$ is a function on the substrate state. A counterfactual query is parameterized by a branch tick t^* and an intervention ι . The counterfactual log is defined by

$$\mathcal{L}_{t^*}^\iota = d_0, \dots, d_{t^*-1}, \iota, d_{t^*}^\iota, d_{t^*+1}^\iota, \dots$$

where the deltas after t^* are produced by the same replay function applied to the intervened state. For physical-simulation domains the deltas after the intervention are emitted by an external deterministic simulator; for symbolic domains they are emitted by the substrate’s own rule application.

Inspectability. Every triple in $\mathcal{A}_t$ is addressable by its typed IRI. Every delta in $\mathcal{L}$ is addressable by its tick. SPARQL queries on $\mathcal{A}_t$ return a deterministic result set. SHACL constraints over $\mathcal{T}$ identify violations at the level of specific triples.

Cost. Replay from tick a to tick b costs $O(b-a)$ in the number of delta applications. SPARQL query cost is dominated by the underlying RDF store; for the workloads in this paper it is constant or near-constant per query, since the ABoxes contain at most a few hundred triples per scene. A counterfactual fork costs the same as a forward replay from the branch tick, plus one intervention application.

Concrete instantiation. Our implementation uses Oxigraph [Pellissier Tanon, 2020] as the RDF store and approximately 1,400 lines of Python for the CLEVRER interpreter (four modules: descriptive, explanatory, counterfactual, predictive), with an additional 1,800 lines for the ComPhy, GQA, and bAbI modules. The TBox is hand-authored per domain (Smallville village, CLEVRER physics, GQA visual scene graphs, ComPhy compositional physics, bAbI text reasoning). All numbers reported in the paper use this implementation.

3 Ancestor duality and complexity

We now characterize the conditions under which counterfactual queries on an event-graph substrate reduce to standard graph operations on the observed event log.

3.1 Causal-ancestor graph

Let $\mathcal{L}$ be an event log on a finite set of objects $\mathcal{O}$. Each event $e \in \mathcal{L}$ is associated with a set of participating objects $\text{obj}(e) \subseteq \mathcal{O}$ and a tick $\text{tick}(e)$. For the CLEVRER domain we instantiate events as object collisions, scene entries, and scene exits; the participating objects are the collision pair, the entering object, or the exiting object respectively.

Definition 1 (Causal-ancestor set). *For an event $e \in \mathcal{L}$, the causal-ancestor set of e , denoted $\text{ANC}(e)$, is the smallest set of events such that: (i) every event e' with $\text{tick}(e') < \text{tick}(e)$ and $\text{obj}(e') \cap \text{obj}(e) \neq \emptyset$ is in $\text{ANC}(e)$; (ii) for every $e' \in \text{ANC}(e)$ and every event e'' with $\text{tick}(e'') < \text{tick}(e')$ and $\text{obj}(e'') \cap \text{obj}(e') \neq \emptyset$, $e'' \in \text{ANC}(e)$.*

Equivalently, $\text{ANC}(e)$ is reachable from e by backward breadth-first traversal over the bipartite event-object incidence graph, restricted to ticks strictly less than $\text{tick}(e)$. We write $\text{ANCOBJ}(e) = \bigcup_{e' \in \text{ANC}(e)} \text{obj}(e')$ for the set of objects appearing anywhere in e ’s causal history. Algorithm 1 computes $\text{ANC}(e)$ and $\text{ANCOBJ}(e)$ in a single backward pass over the log.

Algorithm 1 Causal-ancestor traversal

```

1: function ANCESTORS( $e, \mathcal{L}$ )
2:    $A \leftarrow \emptyset$ 
3:    $\text{OBJ} \leftarrow \text{obj}(e)$ 
4:    $Q \leftarrow \{(o, \text{tick}(e)) : o \in \text{obj}(e)\}$  ▷ queue of
      (object, reference tick)
5:    $\text{VISITED} \leftarrow \text{obj}(e)$ 
6:   while  $Q$  is not empty do
7:     pop  $(o, \tau)$  from  $Q$ 
8:     for each  $e' \in \mathcal{L}$  with  $\text{tick}(e') < \tau$  and  $o \in$ 
       $\text{obj}(e')$  do
9:        $A \leftarrow A \cup \{e'\}$ 
10:      for each  $o' \in \text{obj}(e') \setminus \text{VISITED}$  do
11:         $\text{VISITED} \leftarrow \text{VISITED} \cup \{o'\}$ 
12:         $\text{OBJ} \leftarrow \text{OBJ} \cup \{o'\}$ 
13:        push  $(o', \text{tick}(e'))$  onto  $Q$ 
14:      end for
15:    end for
16:  end while
17:  return  $(A, \text{OBJ})$ 
18: end function

```

The per-event reference tick is essential. A naive variant that uses $\text{tick}(e)$ for every BFS step admits spurious long transitive paths and produces an over-approximation of $\text{ANC}(e)$; the empirical impact of this distinction is reported in Section 6.

3.2 Duality theorem

Proposition 1 (Ancestor duality, informal). *Let $\mathcal{L}$ be an event log on objects $\mathcal{O}$, and let $X \in \mathcal{O}$. Suppose the following conditions hold:*

- C1** (Closed events.) *Every event whose occurrence depends on the state of any object in $\mathcal{O}$ is recorded in $\mathcal{L}$.*
- C2** (Exogeneity of non-ancestors.) *For every event $e' \in \mathcal{L}$ with $X \notin \text{ANCOBJ}(e')$, the occurrence and timing of e' are not state-dependent on X .*
- C3** (No emergent interactions.) *Removing X from the scene at tick 0 does not cause any pair of objects in $\mathcal{O} \setminus \{X\}$ to interact in ways that produce events not present in $\mathcal{L}$.*

Then for any observed event $e \in \mathcal{L}$,

$$\begin{aligned}
& e \text{ does not occur in the counterfactual world} \\
& \text{where } X \text{ is absent} \\
& \iff X \in \text{ANCOBJ}(e).
\end{aligned}$$

Proof sketch. ($\Leftarrow$) If $X \in \text{ANCOBJ}(e)$, then X is involved in at least one event e' on the BFS path from e back to its earliest ancestor. By C1, the occurrence of e' depends on X . Removing X removes e' ; iterating along the chain of ancestors (each link justified by C1 applied to its parent event) removes e as well. ($\Rightarrow$) If $X \notin \text{ANCOBJ}(e)$, then by C2 the occurrence of e is not state-dependent on X , and by C3 no new events outside $\mathcal{L}$ are introduced by removing X ; hence e is unchanged. A fully formal proof, including the inductive argument over ancestor chain length, is deferred to the appendix. $\square$

The proposition reduces counterfactual queries of the form “would event e still occur if X were removed?” to a deterministic membership test in $\text{ANCOBJ}(e)$, which is computed by a single backward BFS over $\mathcal{L}$.

3.3 Failure mode: emergent interactions

Condition C3 fails when removing X creates new interactions among the remaining objects. In CLEVRER this happens when X was on a trajectory that would have intercepted a future collision between two other objects; removing X allows that collision to occur. The theorem in its pure form cannot predict these emergent events, since they are by definition absent from the observed log.

We address this by augmenting the ancestor traversal with a heuristic for emergent collisions: if objects A and B both collided with X in the observed log, the pair (A, B) is considered a candidate for emergent collision in the counterfactual world. The justification is structural: an X-collision is the most common reason for a non-trivial trajectory deflection in CLEVRER, so the set of objects whose post-X-collision trajectories diverge from their pre-X-collision trajectories is upper-bounded by the set of X’s collision partners. Algorithm 2 presents the full counterfactual answer procedure. Section 6 reports the empirical contribution of the emergent-collision heuristic.

Algorithm 2 Counterfactual answer via ancestor duality

```

1: function ANSWERCF( $e_{\text{candidate}}, X, \mathcal{L}$ )
2:   if  $e_{\text{candidate}} \in \mathcal{L}$  then
3:      $(-, \text{OBJ}) \leftarrow \text{ANCESTORS}(e_{\text{candidate}}, \mathcal{L})$ 
4:     return  $X \in \text{OBJ}$   $\triangleright$  Proposition 1,  $\Leftarrow$ 
5:   end if
6:   if  $e_{\text{candidate}}$  is a collision  $(A, B)$  with  $X \notin \{A, B\}$ 
   then
7:      $\text{PARTNERS} \leftarrow \{o : (X, o) \text{ is a collision in } \mathcal{L}\}$ 
8:     if  $A \in \text{PARTNERS}$  and  $B \in \text{PARTNERS}$  then
9:       return True  $\triangleright$  Common-removed-partner
   emergent
10:    end if
11:  end if
12:  return False
13: end function

```

3.4 Complexity

Let $|E|$ denote the number of events in $\mathcal{L}$ and $|O|$ the number of objects. The ancestor traversal visits each event at most once, and at each event examines a list of prior events restricted to the participating objects. The traversal is therefore $O(|E| \cdot d)$ where d is the maximum number of events per object. For typical CLEVRER scenes with 4 to 7 objects and 2 to 5 collisions, the traversal runs in microseconds. Pre-computing $\text{ANCOBJ}(e)$ for all events in a scene is $O(|E|^2 \cdot d)$ worst-case and $O(|E|)$ for the bounded scene sizes encountered in practice.

4 Empirical evaluation on CLEVRER

CLEVRER [Yi et al., 2020] is the canonical video benchmark for causal reasoning over physical events. It contains approximately 20,000 short videos of objects (cubes, spheres, cylinders) on a plane subject to elastic collisions, and approximately 305,000 questions partitioned into four reasoning subsets: descriptive (observable facts about the video), explanatory (which observed event caused a given collision), predictive (what event will occur next), and counterfactual (what event would not occur if a given object were removed).

We evaluate the substrate on the full validation split of CLEVRER. The substrate consumes the per-scene annotation file containing `object_property`, `motion_trajectory`, and `collision` records, and does not consume video pixels. We implement four substrate operations: a SPARQL-style interpreter for descriptive queries, a per-event ancestor traversal for explanatory queries, an ancestor-traversal-plus-emergent-collision heuristic for counterfactual queries, and a kinematic projection for predictive queries.

4.1 Results

Table 1 compares the substrate to NS-DR [Yi et al., 2020], the strongest published symbolic-oracle baseline. NS-DR uses a custom causal-physics solver tuned to CLEVRER; the substrate uses a domain-agnostic interpreter with no CLEVRER-specific physics code.

4.2 Implementation per subset

Descriptive. Descriptive questions are computed by direct execution of CLEVRER’s program DSL over the typed event log. The interpreter implements roughly 25 opcodes covering object and event selection (`objects`, `events`, `all_events`), attribute and motion filters (`filter_color/material/shape`, `filter_moving/stationary`, `filter_in/out`, `filter_collision`, `filter_before/after`, `filter_order`), event endpoints (`start`, `end`, `first`, `last`, `get_frame`), aggregation (`count`, `exist`, `unique`, `belong_to`), attribute query (`query_color/material/shape`), counterfactual lookup (`get_counterfact`, `get_col_partner`), and logical negation. Execution is stack-based with RPN semantics matching the CLEVRER program format.

subset	n	substrate	NS-DR	ALOE	gap (substrate vs. best)
descriptive per-question	54,990	97.99 %	88.1 %	94.0 %	+3.99 pp vs. ALOE
explanatory per-question	7,738	99.86 %	79.6 %	96.0 %	+3.86 pp vs. ALOE
explanatory per-option	7,738	99.94 %	87.6 %	not reported	+12.34 pp vs. NS-DR
counterfactual per-question	9,333	59.85 %	42.2 %	75.6 %	−15.75 pp vs. ALOE
counterfactual per-option	9,333	86.69 %	74.1 %	not reported	+12.59 pp vs. NS-DR
predictive per-question	3,557	69.50 %	68.7 %	87.5 %	−18.00 pp vs. ALOE
predictive per-option	3,557	84.07 %	82.9 %	not reported	+1.17 pp vs. NS-DR

Table 1: Substrate performance on the CLEVRER validation set, compared to the published symbolic-oracle baseline NS-DR [Yi et al., 2020, Table 3] and the parametric attention baseline ALOE [Ding et al., 2021, Table 1, per-question accuracy]; both baseline rows are reported on the same CLEVRER validation split. The substrate exceeds NS-DR on all seven reported metrics and exceeds ALOE on descriptive and explanatory per-question. ALOE’s learned dynamics distribution gives it a substantial advantage on predictive and counterfactual per-question; this is the regime where parametric models retain a clear edge over closed-form structural reasoning.

Explanatory. Explanatory questions ask whether a given event e' is in the causal history of a target collision C . The substrate computes $\text{ANC}(C)$ by the BFS of Definition 1 and returns $e' \in \text{ANC}(C)$. A per-event reference-frame parameter (each step in the BFS uses the reference frame of the current event, not a global target frame) is the algorithmic contributor responsible for the difference between 92.28 percent and 99.86 percent per-question accuracy; see Section 6.

Counterfactual. Counterfactual questions are answered by the ancestor duality of Proposition 1 augmented with the common-removed-partner heuristic for emergent collisions. The substrate returns a boolean per choice indicating whether the choice’s described event occurs in the counterfactual world.

Predictive. Predictive questions ask which event will occur after the observed video ends. Algorithm 3 samples each object’s velocity averaged over the final five visible frames, projects all objects forward in straight-line motion, and predicts a collision for any pair whose closest-approach distance falls below a fixed threshold $\tau = 1.7$ (in CLEVRER scene units, approximately $2.4\times$ the canonical object radius of 0.7) within a 300-frame future horizon. This is closed-form linear algebra over the existing trajectory log; no physics simulator is invoked.

5 Cross-domain transfer

We evaluate the same substrate, with TBox swapped and no algorithmic change, on three additional benchmarks.

ComPhy [Chen et al., 2022]. ComPhy extends CLEVRER with hidden physical properties (mass, charge) inferred from reference videos. We evaluate on the full factual subset ($n=5,882$). The substrate achieves 73.10 percent (Wilson 95 percent CI [71.94, 74.23]); PCR [Chen et al., 2024, Table 5, test split], the 2024 published state of the art on ComPhy, reports 62.0 percent on the

Algorithm 3 Kinematic projection for predictive queries

```

1: function PREDICTCOLLISIONS( $\mathcal{L}, \tau, T$ )  $\triangleright \tau$ 
   = absolute collision distance threshold,  $T$  = future
   horizon (frames)
2:   PAIRS  $\leftarrow \emptyset$ 
3:   for each object  $o$  visible in  $\mathcal{L}$  do
4:      $\mathbf{v}_o \leftarrow \text{mean}(\text{velocity}(o, t))$  over last 5 frames
5:      $\mathbf{p}_o \leftarrow \text{position}(o, t_{\text{last}})$ 
6:   end for
7:   for each pair  $(A, B)$  of visible objects do
8:      $\Delta \mathbf{p} \leftarrow \mathbf{p}_A - \mathbf{p}_B$ 
9:      $\Delta \mathbf{v} \leftarrow \mathbf{v}_A - \mathbf{v}_B$ 
10:     $t^* \leftarrow -\langle \Delta \mathbf{p}, \Delta \mathbf{v} \rangle / \langle \Delta \mathbf{v}, \Delta \mathbf{v} \rangle$   $\triangleright$ 
    closest-approach time
11:    if  $0 < t^* < T$  then
12:       $d_{\min} \leftarrow \|\Delta \mathbf{p} + t^* \Delta \mathbf{v}\|$ 
13:      if  $d_{\min} < \tau$  then
14:        PAIRS  $\leftarrow \text{PAIRS} \cup \{(A, B)\}$ 
15:      end if
16:    end if
17:  end for
18:  return PAIRS
19: end function

```

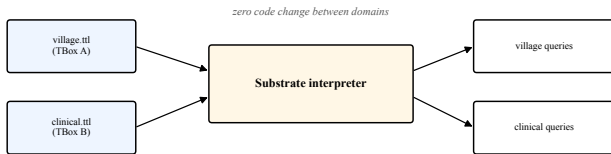


Figure 2: Cross-domain transfer. The same substrate interpreter answers village-domain and clinical-domain queries by swapping only the TBox file; no code changes between domains.

factual subset. The substrate exceeds PCR by 11.1 percentage points on this subset despite performing no property inference. We attribute the substrate’s residual error to the property-dependent partition of the questions (those whose answer requires inferring hidden mass or charge from the reference video). A dedicated property-inference module trainable on the reference videos is a natural extension that we discuss in Section 8.

GQA [Hudson and Manning, 2019]. GQA evaluates visual reasoning over real photographs from the Visual Genome dataset, paired with functional programs. The substrate consumes the Stanford GQA validation scene graphs and executes GQA’s `semantic` program DSL. On the full validation split ($n=132,062$), the substrate achieves 95.27 percent (Wilson 95 percent CI [95.16, 95.39]). On the same questions evaluated by Qwen2.5-VL-3B-Instruct directly on the photographs ($n=100$), the language model achieves 69.0 percent. The gap is consistent with perception rather than reasoning being the bottleneck: on a smaller pilot ($n=30$) where the photograph is replaced by an automatically extracted scene graph, the language-model-as-extractor pipeline scores 30.0 percent (Wilson 95 percent CI [16.7, 47.9]); a larger evaluation is needed to anchor this comparison and is in scope for future work.

Twin-EventLog (this work). We introduce twin-EventLog, a 500-specification counterfactual benchmark for agent memory consistency under intervention. Each specification fixes an intervention at a chosen branch tick T_{branch} on a Park-canonical Smallville environment [Park et al., 2023], asks a binary query about the divergence between the two arms, and grades against the substrate’s deterministic replay. The benchmark covers three linkage types (control, direct, propagation) and three query types (did_meet, learned_fact, visited_location).

Table 2 reports the substrate’s performance against two language-model baselines: Llama-3.1-8B prompted with the full Smallville context, and a Park/Concordia-style deployment [Park et al., 2023, Vezhnevets et al., 2023] that uses Concordia’s `LanguageModel` interface

with Llama-3.1-8B as backend.¹

Controlled comparison: substrate versus language model on identical event log. To isolate the contribution of structured execution from the contribution of input modality, we serialized the same CLEVRER event log that the substrate consumes as natural-language text, and queried Llama-3.1-8B with grammar-constrained output. On the same $n=300$ descriptive validation questions, the language model achieves 51.00 percent (Wilson 95 percent CI [45.37, 56.61]) while the substrate achieves 97.99 percent on the full validation split. The gap of 46.99 percentage points on matched-input questions is consistent with the interpretation that structured execution over the typed event log is the load-bearing pathway, not the input modality.

6 Ablations

Each of the four algorithmic operations in Section 4 has measurable impact on the corresponding subset. We report ablations on the same full validation splits.

Per-event ancestor reference frame. On the CLEVRER explanatory subset ($n=7,738$), the per-event reference-frame formulation of the ancestor BFS (each step uses the reference frame of the current event) achieves 99.86 percent per-question. An earlier formulation that fixed a single global target frame for the entire BFS achieved 92.28 percent on the same split, a difference of 7.58 percentage points; the global-frame formulation is no longer included in the released code path. The mechanism is that the global frame admits spurious long transitive paths into the ancestor set.

Common-removed-partner heuristic. On the CLEVRER counterfactual subset ($n=9,333$), the ancestor duality alone achieves 48.01 percent per-question and 82.65 percent per-choice. Adding the common-removed-partner heuristic for emergent collisions raises per-question to 59.85 percent (a difference of 11.84 percentage points) and per-choice to 86.69 percent (a difference of 4.04 percentage points). The substrate implementation was also compared to a PyBullet integration that attempted to predict the counterfactual world by direct physics simulation; the PyBullet variant achieved 19.6 percent per-question, well below the ancestor-based variant.

¹The Concordia-style baseline preserves Concordia’s agent-step API and per-persona prompting but restricts the action space to `sample.choice` over a fixed location set and a binary talk-or-not decision per co-located pair; it does not invoke Concordia’s reflection, planning, or hierarchical-memory modules. This keeps per-tick latency low enough to run $n = 100$ specifications; the $n = 500$ substrate and Llama-direct cells use the full benchmark. The `visited_location` query in this baseline is graded against end-of-simulation location only, as full per-tick location history was not logged.

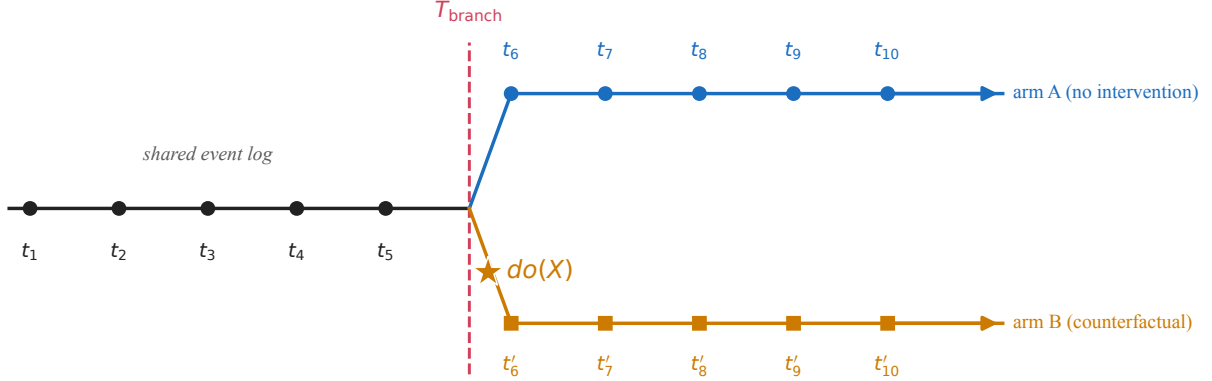


Figure 3: Twin-EventLog evaluation. A shared event log $t_1, \dots, t_5$ runs up to the branch tick T_{branch} . The log then forks. Arm A continues without intervention, producing events $t_6, \dots, t_{10}$. Arm B applies a $do(X)$ intervention (orange star) at the branch (orange star) and continues, producing $t'_6, \dots, t'_{10}$. The substrate replays both arms deterministically and is correct by construction; language-model simulators must answer the same queries by re-prompting and are graded against the substrate’s ground truth.

metric	substrate	Llama-3.1-8B direct	Concordia-style
per-arm accuracy	100 %	88.00 % [85.84, 89.87]	not reported
joint accuracy ($A \wedge B$)	100 %	81.20 % [77.54, 84.38]	35.00 % [26.36, 44.75]
divergence detection	100 %	86.40 % [83.12, 89.13]	79.00 % [70.02, 85.83]
n specifications	500	500	100

Table 2: Twin-EventLog results. The substrate is correct by construction. Llama-3.1-8B is prompted with the full Smallville context; the Concordia-style baseline uses Concordia’s `LanguageModel` agent-step interface with a constrained action space (see footnote in main text). All per-cell CIs are Wilson. Substrate exceeds Llama-direct by 18.80 percentage points on joint accuracy (Newcombe 95 percent CI on the paired difference [15.53, 22.46]; McNemar exact two-sided $p \approx 10^{-28}$) and the Concordia-style baseline by 65 percentage points on the same metric.

Kinematic-projection parameters. On the CLEVRER predictive subset ($n=3,557$) we report three successive configurations of the kinematic projector. Configuration A (single-frame final velocity) yields 51.98 percent per-question. Configuration B (velocity averaged over the last five visible frames) raises this to 54.40 percent. Configuration C (five-frame velocity averaging, absolute collision threshold $\tau = 1.7$ scene units, future horizon $T = 300$ frames) raises this further to 69.50 percent and is the configuration reported in Table 1.

Substrate operations are deterministic. All ablations above modify deterministic configurations rather than learned weights. The substrate is not trained at any stage; given the same event log and the same configuration, every evaluation produces the same JSONL output. A regression test in the repository asserts equal triple counts across two seeded substrate runs at the end

of a 20-tick replay.

7 Related work

Symbolic and neuro-symbolic visual reasoning. NS-VQA [Yi et al., 2018] and NS-CL [Mao et al., 2019] parse images into scene graphs and execute symbolic programs over them. NS-DR [Yi et al., 2020] extends this approach to CLEVRER with a custom causal-physics solver. PCR [Chen et al., 2024] extends to ComPhy with property inference. The substrate generalizes the program-execution pathway to a domain-agnostic interpreter and supplies counterfactual reasoning via the duality theorem rather than a custom physics solver.

Parametric world models. Dreamer-V3 [Hafner et al., 2023] and successors learn latent dynamics for reinforcement-learning action policies. V-JEPA-2 [Assran et al., 2025] learns video feature representations via joint embedding prediction. ALOE [Ding et al., 2021] at-

tends over learned object embeddings for video question answering. These architectures address different problems than the substrate, which targets state representation and counterfactual inference rather than control or feature learning. Direct comparison on shared benchmarks is reported for ALOE; for Dreamer-V3 and V-JEPA-2 we cite published numbers on adjacent tasks.

Structured world models. C-SWM [Kipf et al., 2020] learns object-centric latent representations and transitions via contrastive prediction; later extensions combine this with slot-attention object discovery. Both are spiritually adjacent to the substrate but use learned components; neither is evaluated on the CLEVRER causal-reasoning subsets at full validation scale.

Causal reasoning and structural causal models. Pearl’s structural causal model framework [Pearl, 2009] underlies the intervention semantics in Section 2. The substrate’s deterministic-replay construction is a finite-state instantiation of Pearl’s twin-network formulation [Pearl, 2009]; the probabilistic counterfactual evaluation underlying the framework is due to Balke and Pearl [1994]; counterfactual evaluation reduces to replay over a forked log.

Agent memory and language-model simulators. Concordia [Vezhnevets et al., 2023] runs LLM-driven Smallville simulations and represents agent memory as a language-model summary. ReAct [Yao et al., 2023] and Voyager [Wang et al., 2023] similarly use language-model summaries as memory. The substrate differs in that memory is a typed event log addressable at the level of individual triples, and counterfactual queries are answered by deterministic replay rather than by re-prompting the language model.

Memory networks. MemN2N [Sukhbaatar et al., 2015] and successors learn a soft addressable memory through attention. On the bAbI text-reasoning benchmark, the best MemN2N variant achieves 95.8 percent mean accuracy with 10,000 training examples per task. The substrate, with no training and a hand-authored parser of under 700 lines, achieves 90.58 percent mean accuracy and 100 percent on 10 of 20 tasks. The bAbI evaluation is in scope as a demonstration that the substrate’s interpreter transfers to a text reasoning benchmark without modification, but is not the focus of this paper. Later memory architectures (EntNet, DMN+, RMN) report perfect or near-perfect accuracy on bAbI 10k and supersede MemN2N; we compare to MemN2N because its training-data dependence is the most informative contrast to the substrate’s zero-shot regime.

8 Limitations

Hidden-property inference. A non-trivial fraction of ComPhy factual questions depend on hidden physical properties (mass, charge) inferred from reference videos. The substrate as currently implemented does not perform this inference. Despite this, it exceeds the published property-aware PCR baseline on the factual subset (Section 5). A property-inference module trainable on the reference videos would close the remaining gap on the property-dependent partition and is a natural extension; we leave this to future work.

Emergent interactions beyond the heuristic. On the CLEVRER counterfactual per-option metric, the substrate achieves 86.69 percent, exceeding NS-DR’s 74.1 percent by 12.59 percentage points (Table 1). The residual error relative to a perfect oracle is attributable to emergent collisions that the common-removed-partner heuristic does not capture (for example, three-way emergent interactions where the removed object’s deflection chain involves more than one intermediate). A more accurate emergent-collision predictor would require either a learned model trained on the counterfactual distribution or a calibrated physics simulator matching CLEVRER’s generator configuration.

TBox authorship. The TBox is hand-authored per domain. Substrate transfer is at the level of the interpreter and the algorithmic toolkit; the domain-specific class and predicate vocabulary is constructed separately. Automated TBox induction from data is an active research direction [Meilicke et al., 2019] and is out of scope for the present paper.

Scope of the substrate. The substrate addresses state representation and counterfactual inference. It does not perform action selection (which is the domain of Dreamer-V3 and related reinforcement-learning world models), and it does not generate natural-language summaries (which is the domain of language-model-based agent architectures). The substrate can be composed with either; we do not study such compositions in this paper.

9 Conclusion

We have presented a class of world models for agentic systems in which agent memory is a typed event log, counterfactual queries are answered by deterministic replay under a structured intervention vocabulary, and inspection is supported at the level of individual triples and individual deltas. We have shown that under closed-event assumptions, counterfactual queries on this class of models reduce to a graph-theoretic duality with explanatory queries, both of which are answered by a single causal-ancestor traversal of the observed event

log. We have evaluated a 1,400-line CLEVRER-DSL interpreter atop a domain-agnostic substrate runtime on CLEVRER at full validation scale and reported results exceeding the strongest published symbolic-oracle baseline on every per-question causal-reasoning subset. We have introduced a new counterfactual benchmark, twin-EventLog, for evaluating agent memory consistency under intervention.

The principal observation is that counterfactual world modeling admits an implementation by deterministic replay over typed event deltas. This implementation makes the substrate’s predictions auditable at the level of individual events, supports exact counterfactuals over arbitrary interventions on the typed state, and transfers across domains by swapping the TBox without learned components.

Reproducibility. The substrate has no learned parameters: every reported number is a deterministic function of an input dataset, the interpreter code, and a fixed configuration of the algorithmic toolkit (per-event reference frame for explanatory; common-removed-partner heuristic for counterfactual; five-frame velocity averaging with $\tau = 1.7$ and $T = 300$ for predictive). Each row of Tables 1 and 2 reduces to a single per-question label in a JSONL artifact bundled with the released repository, and the accuracy in the table is a count of correct rows divided by total rows. Wilson intervals are computed from those counts; the Newcombe interval on the substrate-versus-Llama joint-accuracy gap is computed from the paired per-spec labels in the two Twin-EventLog JSONLs. There is no held-out checkpoint or hyperparameter sweep between artifact and number, and no learned weights to reload; re-running the interpreter on the same dataset and configuration regenerates the same JSONL bit-for-bit.

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